Espiner — Bag Strength Test Development

Bag Clamp Fixture | Design Session Progress Report

PREPARED BY
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PERIOD | MACHINE
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CONFIDENTIAL · ISO 13485:2016 · ALL MATERIALS SS 316 THROUGHOUT · FULL MATERIAL CERTS REQUIRED

A full design session completed — the mechanical clamping fixture holding a ripstop nylon conical bag during destructive compression testing at 590 N. All component geometries are defined, key design decisions locked, ready for Fusion 360 CAD finalisation. The one primary measurement task remains before machining release.

ACCOMPLISHMENTS THIS SESSION

- **Bag geometry confirmed** — Clamp plane Ø67.8 mm, upper ref Ø73.2 mm, taper ~8.1° half-angle
- **Clamping principle finalised** — Conical SS surfaces + 1 mm Shore 20A silicone liners. Dual grip: threaded rod (torque-controlled) + self-energising wedge at 8.1° self-locking boundary
- **All 7 component dimensions defined** — Bottom plate, standoffs, top plate, split collar, support ring, silicone liners, closing rods
- **Structural calculations verified** — All safety factors confirmed (see table)
- **5 documents produced** — ENG-CLJIG-001 Rev E, weekly report, CAD dimensions ref, ring cross-sections, this summary

KEY DESIGN DECISIONS LOCKED

Dovetail Retention
Through male dovetail rail (50 × 12 mm). Collar slides into plate — shoulders resist uplift at bag failure. Zero retention fasteners.

Leadscrew Closing
Half A = fixed jaw. Half B driven by ×2 M8 rods. Guided by dovetail walls — cannot skew regardless of tightening sequence.

Half A Adjustability
M6×0.75 fine-pitch screw + slotted lock bolts. Bore centreline alignable with machine axis. Position recorded in DHF per session.

Bore Surface Finish
Ra 3.2–6.3 µm — deliberate roughness for silicone RTV mechanical key. As-machined, no finishing pass — reduces cost.

Base Plate Fixing
5-hole pattern: 4× #10-32 UNF + 1× ½-20 UNC. Countersinks on top face. No Mark-10 modification required.

Elastomer Cone — Rejected
Cone on TP to stabilise ring — rejected. Ring already stabilised by clamped bag wall. Revisit after first test data only.

PENDING BEFORE MACHINING RELEASE

ITEM	ACTION	PRIORITY
Bag taper angle	Measure 5+ bags at 4+ heights — all dims depend on this	BLOCKER
Top plate dims	Confirm rectangle dimensions in Fusion 360	CAD
Half A fixing holes	Confirm positions in CAD — clear of dovetail & rod holes	CAD
M8 rod hole positions	Min 12 mm clearance above/	CAD

ITEM	ACTION	PRIORITY
	below bore edges	
Standoff bolt circle	Maximise spread within plate footprint	CAD
Clamping torque protocol	Establish from first test — suggested 5 Nm per rod	TEST
Rod & fastener lengths	Calculate from assembled CAD model	CAD

9 hours effort — 5 hrs CAD + 4 hrs design development

STRUCTURAL CALCULATIONS VERIFIED

CHECK	RESULT	SF	
Standoff stiffness	k = 2,897 N/mm δ = 0.19 mm at 560 N	—	✓
Standoff fatigue	14.1 MPa bending vs 120 MPa endurance	8.5×	✓
Ring wall (min section)	3.4 mm wall, thick-walled cylinder	9.5×	✓
Friction grip	697 N grip > 590 N test load	>1×	✓
Dovetail root stub	16 mm — adequate	—	✓
Assembly clearance	2.5 mm each side — elastomer through bore	—	✓

IMMEDIATE NEXT STEP

Measure 5+ bag samples at 4+ heights each